A.K. Vishwanathan

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Education

Indian Institute of Technology Kharagpur *Dec 2020 - Apr 2025*

Integrated B.Sc. (Honours) and M.Sc. in Chemistry; CGPA: 7.59/10

Publications

iToFD: An automated unsupervised framework for weld defect detection and measurement using Time-of-Flight Diffraction data

AK Vishwanathan, A. Dutta, A. Mukherjee, S. K. Singh, and Surjya K. Pal *2025*

ASME Journal of Nondestructive Evaluation, Diagnostics and Prognostics of Engineering Systems

A method and system for automatically detecting weld defects in ultrasonic Time-of-Flight Diffraction (ToFD)

AK Vishwanathan, A. Dutta, A. Mukherjee, and Surjya K. Pal

Indian patent **granted** by The Patent Office, Government of India (Patent No. 529585) *2024*

Work Experience

Eve Inc. | Software Engineer – Machine Learning *May 2025 - Present*

Four Vectors: Egocentric Video Pipelines

- Architected a **Backblaze B2–RunPod** GPU pipeline processing **111.6 h** of raw tapes at **64%** yield
- Caught near-duplicate and train/test-leaked episodes by pooling 384-d **DINOv2** embeddings of 8 frames into a **FAISS** cosine index (top-5 search), backed by 64-bit perceptual hashes for re-encodes
- Fused **optical flow** and **YAMNet** audio for idle time; redacted PII via **YOLO-World** and **SAM 2**

Ode: Real-Time Recommendation Systems

- Owned the **embedding pipeline** for a real-time event recommender (**50K** events in **Pinecone**); trained a **two-tower model** on 500K user–event interactions, improving **recall@50** by **12%**
- Cut **p95** latency from **400 to 150 ms** at peak; served it as a Dockerised **Flask** API on **AWS EC2**
- Scaled nearline fine-tuning from a single EC2 instance to **SageMaker**, with a Glue → S3 **ETL pipeline** and **blue-green** index swaps orchestrated via **Step Functions** and **Lambda**

Research Experience & Internship

Mechanistic Interpretability of Text-to-Image Encoders *Jul 2026 - Sep 2026*

Mentor: Michael Toker (PhD researcher, Technion) *Independent research collaboration*

- Adapted **Diffusion Lens** and **information-flow** methods to **FLUX.2 Klein (4B)**'s **Qwen3** encoder
- Built three complementary **interpretability lenses** (logit, diffusion, Jacobian) and fused per-token emergence layers into token cards classifying concept formation into **seven distinct patterns**
- Diagnosed a silent **activation-cache corruption** (RMSNorm writing through aliased views under `accelerate` CPU offload) and rebuilt capture with fingerprint checks for **bit-reproducible** runs

Training and Behaviour of Optical Neural Networks (ONN) *Jul 2023 - Dec 2023*

Guides: Prof. Dirk Englund, Dr. Vadlamani (postdoc) *EECS, Massachusetts Institute of Technology*

- Built a **loss-based data-pruning** pipeline that ranks training examples by cross-entropy and keeps training only on the hardest top-N subset, for both optical and electronic classifiers

- Implemented a 2-layer 256-mode **Clements Mach–Zehnder** ONN (∼132k parameters); trained on only the hardest 20k and 10k MNIST examples, it held **97.67%** and **97.01%** test accuracy
- Designed an **iterative pruner**; an MLP held **96.97%** accuracy on only **6.4k of 60k** MNIST images

ML-Based Image Analysis of ToFD Data *Sep 2022 - Apr 2025*
Guide: Prof. Surjya K. Pal *CoEAMT, Indian Institute of Technology Kharagpur*

- As technical lead on a ∼USD 60k government-funded **AI4ICPS** project, architected the patented **unsupervised B-scan pipeline**: Hilbert-envelope denoising and **DBSCAN** defect clustering
- Extended it with a supervised **LSTM/GRU** model that predicts adaptive per-A-scan thresholds from windowed features, reaching ∼**0.91** validation R^2 against manually annotated B-scans
- **Parallelised** A-scan processing across CPU cores and shipped the desktop app **iToFD**, cutting analysis of a 15 m weld from **8–10 hours** of manual inspection to **2–3 minutes** at **95% accuracy**

Multi-object Goal Navigation *Jun 2023 - Aug 2023*
Guide: Prof. Mohan Sridharan *Computer Science, University of Birmingham*

- Simulated **multi-object goal navigation** in **AI Habitat** on the **Gibson** dataset, studying trajectory optimisation for agents localising several object classes under partial observability
- Analysed **DRQ**, **Spatial Relation Graph + GCN** and **sequence-agnostic** navigation, assessing how spatial priors and semantic mapping cut search cost in multi-room environments

Bilevel Hyperparameter Optimisation *Dec 2022 - Jan 2023*
Guide: Prof. Ankur Sinha *Centre for Data Science & AI, Indian Institute of Management Ahmedabad*

- Framed neural-network **hyperparameter tuning** as a **bilevel program** (validation loss over architecture and regularisation; training loss over weights), solved by a **micro-genetic algorithm**
- Added a **linear-program** step computing steepest-descent directions for hyper-local refinement; on **MNIST** and **CIFAR-10** it lifted the algorithm's test accuracy from **73%** to **78%**

Key Courses & Certifications

- **Computer Science:** Programming and Data Structures, Artificial Intelligence: Foundations and Applications, Artificial Intelligence for Manufacturing, Dependable and Secure AI–ML
- **Mathematics:** Linear Algebra, Probability and Statistics, Numerical and Complex Analysis, Transform Calculus, Advanced Calculus
- **Certifications (Andrew Ng):** Machine Learning (Stanford); Neural Networks and Deep Learning; Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization

Technical Skills

Languages	Python, SQL, C++, C, MATLAB, LaTeX
Libraries	PyTorch, TensorFlow, Keras, HuggingFace (Transformers, Diffusers, Accelerate), NumPy, Pandas, OpenCV, MediaPipe, Matplotlib, FAISS, Flask, Pinecone
Tools	AWS (SageMaker, Glue, S3, Step Functions, Lambda, EC2), Docker, Git

Honours & Presentations

- **Gold medal** (team) at **Inter IIT Tech Meet 12.0** for AI-enabled ToFD weld defect detection *(2023)*
- Presented AI-enabled ToFD software at **Tata Steel Technical Exhibition**, Jamshedpur *(2024)*
- Achieved an All-India Rank of **5287** in **JEE Advanced** out of **250,000** eligible candidates *(2020)*
- Scored **96.11 percentile** in **JEE Main**, an exam with over **1.1 million** registered candidates *(2020)*